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Large-scale surface hopping simulation of charge transport in hexagonal molecular crystals: role of electronic coupling signs

Zirui Wang¹, Jiawei Dong¹ and Linjun Wang^{*} 

Key Laboratory of Excited-State Materials of Zhejiang Province, Department of Chemistry, Zhejiang University, Hangzhou 310058, People's Republic of China

E-mail: ljwang@zju.edu.cn

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Abstract

In large-scale surface hopping simulations with a huge number of electronic states, trivial crossings could easily lead to incorrect long-range charge transfer and induce large numerical errors. We here study the charge transport in two-dimensional hexagonal molecular crystals with a parameter-free full crossing corrected global flux surface hopping method. Fast time-step size convergence and system size independence have been realized in large systems containing thousands of molecular sites. In hexagonal systems, each molecular site has six nearest neighbours. We find that the signs of their electronic couplings have a strong impact on the charge mobility and delocalization strength. In particular, changing the signs of electronic couplings can even lead to a transition from hopping to band-like transport. In comparison, such phenomena cannot be observed in extensively studied two-dimensional square systems. This is attributed to symmetry of the electronic Hamiltonian and distribution of the energy levels. Due to its high performance, the proposed approach is promising to be applied to more realistic and complex systems for molecular design.

Keywords: large-scale surface hopping, charge transport, hexagonal molecular crystals, electronic coupling

(Some figures may appear in colour only in the online journal)

1. Introduction

In chemistry, physics, biology and materials science, many important phenomena involve strong electron-nuclei couplings and belong to the category of nonadiabatic dynamics, where the Born–Oppenheimer approximation breaks down. As a popular mixed quantum–classical dynamics method to describe these nonadiabatic processes, trajectory

surface hopping (TSH) [1] makes stochastic hops among the adiabatic potential energy surfaces. Along each trajectory, the nuclear motion follows the Newtonian equation, while the electronic wave function propagates through the time-dependent Schrödinger equation (TDSE). Combined with *ab initio* electronic structure calculations, TSH simulations have realized a good balance between accuracy and efficiency as demonstrated in many applications to study photoisomerization [2, 3], proton transfer [4, 5], charge transport [6, 7], exciton diffusion [8, 9], singlet fission [10, 11] and various other dynamical processes [12–16].

¹ These authors contributed equally to this work.

^{*} Author to whom any correspondence should be addressed.

Despite the popularity of TSH methods, there are still some bottlenecks for nonadiabatic dynamics simulation of large-scale systems with a large number of electronic states. Trivial crossings [17, 18] happen between two states when their electronic coupling is weak and the nonadiabatic coupling (NAC) shows a sharp peak within a small time interval. The resulting transition probability changes quickly with time and thus is hard to describe efficiently in numerical simulations. In fact, trivial crossings easily lead to incorrect long-range charge transfer, introducing significant overestimation of the charge displacement [19, 20]. To avoid mistakes in dealing with trivial crossings, the strategy of using adaptive time-step sizes has been imported naturally. The general idea is adopting decreased time-step size at detected trivial crossing points [21–23]. Theoretically, as long as the time-step size is small enough, each surface crossing can be accurately described. However, this means huge sacrifice of the efficiency when studying extended systems with a large number of electronic states. In principle, we may also try to identify all relevant trivial crossing regions, rearrange the corresponding electronic wave function coefficients, and determine surface hops manually. Fabiano *et al* [24] implemented a maximum ‘overlap’ procedure and generated an index mapping of the old states to the new ones. Fernandez-Alberti *et al* [17] also presented a min-cost algorithm to identify trivial crossings through tracking the states in time. Recently, the full crossing corrected global flux surface hopping (FCC-GFSH) method has been developed by Wang and coworkers [25, 26]. Compared with the traditional FSSH, GFSH [27] defines the hopping probabilities purely based on the population flux between adiabatic states without involving explicitly NACs, thus solving the numerical instability of the surface hops at trivial crossings, and better describing the superexchange mechanism of population transfer. In addition, the surface crossings are classified by state tracking and each crossing is treated accordingly. As demonstrated in a series of one- and two-dimensional systems, FCC-GFSH has shown high performance in giving accurate dynamics for complex trivial crossings within a short time interval, which makes it suitable for large-scale nonadiabatic dynamics simulations of general systems.

In many molecular crystals, the molecules are arranged in layers and adopt the herringbone packing [28–30]. We here consider general two-dimensional hexagonal molecular crystals and adopt the Holstein-type model with open boundary condition to carry out nonadiabatic dynamics simulations of charge transport. Note that Fratini *et al* [31] computed the localization characteristics of molecular crystals with a large set of electronic couplings between neighbouring molecules, which demonstrated distinct mobilities as the sign of electronic coupling is changed. In the present study, we clarify the dynamics dependence on the electronic coupling signs and obtain a detailed understanding of the mechanism of charge transport in these more realistic molecular semiconductors.

2. Methodology

2.1. Hamiltonian

As showed in figure 1, we consider the hexagonal lattice to describe molecular crystals. For simplicity, each molecular site is surrounded by six adjacent sites with equal spacing L . In order to systematically study the charge transport, we construct a general Holstein-type model Hamiltonian, $\hat{H}(\mathbf{r}; \mathbf{R})$, with N molecular sites, where $\mathbf{r}$ and $\mathbf{R}$ are the coordinates of electronic and nuclear degrees of freedom. The total Hamiltonian consists of two parts, $\hat{H} = \hat{H}_e + \hat{H}_n$, which are respectively expressed as

$$\begin{aligned} \hat{H}_e = & \sum_{i=1}^N (\varepsilon_i + \alpha R_i) |\chi_i\rangle \langle \chi_i| \\ & + \sum_{i=1}^N V_a (|\chi_i\rangle \langle \chi_{i_a}| + |\chi_{i_a}\rangle \langle \chi_i|) \\ & + \sum_{i=1}^N V_b (|\chi_i\rangle \langle \chi_{i_b}| + |\chi_{i_b}\rangle \langle \chi_i|) \\ & + \sum_{i=1}^N V_c (|\chi_i\rangle \langle \chi_{i_c}| + |\chi_{i_c}\rangle \langle \chi_i|), \end{aligned} \quad (1)$$

$$\hat{H}_n = \sum_{i=1}^N \frac{1}{2} \left(K R_i^2 + \frac{P_i^2}{m} \right). \quad (2)$$

Here, each site i is associated with a diabatic orbital $|\chi_i\rangle$ and a harmonic vibrational degree of freedom R_i with the corresponding momentum P_i , the same force constant K and effective mass m . α is the local electron–phonon coupling constant. Due to the translational symmetry, we consider only three independent electronic couplings V_a , V_b , and V_c in the three directions a , b , and c to describe the nearest neighbour interactions. i_a (i_b , i_c) is the index of the considered nearest neighbour of site i along the a (b , c) direction. At the equilibrium geometry, the onsite energies, ε_i , of all molecular sites are the same and set to zero. The open boundary condition is considered. Agreeing with previous studies [25, 26], the parameters are chosen as $\alpha = 35\,000\text{ cm}^{-1}\text{ nm}^{-1}$, $K = 14\,500\text{ u ps}^{-2}$, $m = 250\text{ u}$, and $L = 0.5\text{ nm}$.

The charge transport simulation is carried out with a homemade software package Simulation Package for non-Adiabatic Dynamics in Extended systems (SPADE), which supports the use of arbitrary function forms to construct the diabatic Hamiltonian and state-of-the-art surface hopping methods to study nonadiabatic dynamics. The FCC-GFSH algorithm [25, 26] is adopted to treat the complex surface crossings. A series of independent surface hopping trajectories is generated to give a reliable description of the time-dependent dynamics. Here, only a brief overview of the key algorithms is given below.

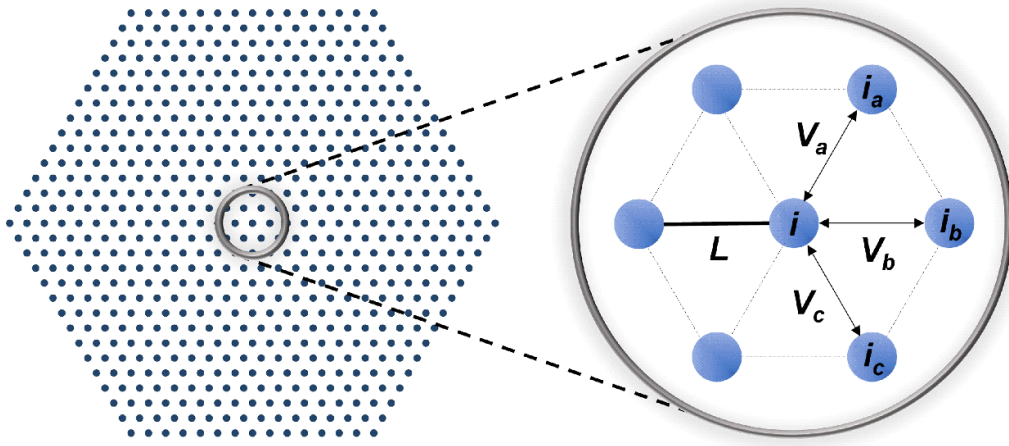


Figure 1. Graphical representation of two-dimensional hexagonal molecular crystals. For simplicity, each blue solid circle represents a molecular site, which is associated with a diabatic state and couples with six first-neighbour molecular sites. The distance between all adjacent molecules is uniformly set as $L = 0.5$ nm. The charge is initially located at the centre of the system. V_a , V_b and V_c are the electronic couplings along the three indicated directions for each site i .

2.2. Initial conditions

For each trajectory, the initial electronic wave function, $\psi(\mathbf{r}, 0)$, nuclear coordinates, $\mathbf{R}$, and momenta, $\mathbf{P}$, are initialized as needed. As an orthogonal diabatic basis $\{|\chi_i(\mathbf{r})\rangle\}$ has been defined, the initial electronic wave function can be expressed as

$$|\psi(\mathbf{r}, 0)\rangle = \sum_i c_i(0) |\chi_i(\mathbf{r})\rangle. \quad (3)$$

Here, $\{c_i(0)\}$ are the expansion coefficients at time zero. To efficiently simulate the diffusion process, the charge is initially placed at the centre of the system. Thereby, we set $c_i(0) = \delta_{io}$, where o is the index of the central molecular site. For harmonic oscillators, the initial nuclear coordinates, $\mathbf{R}$, and momenta, $\mathbf{P}$, are sampled via the Wigner distribution [32],

$$W(\mathbf{R}, \mathbf{P}) = \frac{1}{(\pi \hbar)^N} \prod_i \exp\left(-\frac{(R_i - R_{i0})^2}{2\sigma_{R_i}^2}\right) \exp\left(-\frac{P_i^2}{2\sigma_{P_i}^2}\right). \quad (4)$$

Here, R_i and P_i are the coordinate and momentum of the i th classical degree of freedom with variances σ_{R_i} and σ_{P_i} , respectively. R_{i0} is the corresponding equilibrium coordinate. At a finite temperature T , the two variances are set as [6, 33]

$$\sigma_{R_i}^2 = \frac{\hbar}{2m\omega} \coth\left(\frac{\hbar\omega}{2k_B T}\right), \quad (5)$$

$$\sigma_{P_i}^2 = \frac{\hbar m\omega}{2} \coth\left(\frac{\hbar\omega}{2k_B T}\right), \quad (6)$$

where $\omega = (K/m)^{1/2}$ is the vibrational frequency and k_B is the Boltzmann constant. It is worth mentioning that in the

presence of feedback from the charge, the equilibrium nuclear coordinates should deviate from zero. Hence, after injecting the charge, an energy relaxation process is performed to minimize the system energy and $\{R_{i0}\}$ are reassigned before the sampling of nuclear coordinates [25, 26].

At a given geometry, $\mathbf{R}$, diagonalizing the diabatic Hamiltonian, $\hat{H}(\mathbf{r}; \mathbf{R})$, we obtain all the eigenstates $\{|\phi_i(\mathbf{r}; \mathbf{R})\rangle\}$, eigenenergies $\{E_i(\mathbf{R})\}$, and the corresponding transformation matrix $\mathbf{U}(\mathbf{R})$, whose elements are defined as $U_{ji}(\mathbf{R}) = \langle\chi_j(\mathbf{r})|\phi_i(\mathbf{r}; \mathbf{R})\rangle$. The wave function can also be expressed in the adiabatic representation as

$$|\psi(\mathbf{r}, t)\rangle = \sum_i w_i(t) |\phi_i(\mathbf{r})\rangle. \quad (7)$$

Here, $\{w_i\}$ are the expansion coefficients, which can be computed through the representation transformation,

$$w_i(t) = \sum_j c_j(t) U_{ji}^*(\mathbf{R}(t)). \quad (8)$$

To ensure that the charge is initially placed closed to the central molecular site, the initial active state, a , is chosen as the adiabatic state with the maximum charge population, i.e. $a = \text{argmax}_i (|w_i(0)|^2)$.

2.3. Nuclear and electronic time evolution

In surface hopping, the nuclei move on the active state according to the Newtonian equation. To characterize the system-bath interaction, the nuclear motion is described by the Langevin equation [6, 34]. The force on the i th classical degree of freedom at time t is

$$F_i(t) = -\frac{\partial E_a(\mathbf{R}(t))}{\partial \mathbf{R}_i(t)} - \gamma m \dot{\mathbf{R}}_i(t) + \xi, \quad (9)$$

where $E_a(\mathbf{R}(t))$ is energy of the active state, γ is the friction coefficient, and ξ is a stochastic force with zero mean value and standard deviation of $(2\gamma m k_B T / \Delta t)^{1/2}$. For every time step, equation (9) is solved numerically by the fourth-order Runge–Kutta algorithm with a time-step size of Δt [22].

The electronic wave function is propagated through the TDSE,

$$i\hbar \frac{\partial |\psi(\mathbf{r}, t)\rangle}{\partial t} = \hat{H}(\mathbf{r}; \mathbf{R}(t)) |\psi(\mathbf{r}, t)\rangle. \quad (10)$$

We adopt the locally diabatic (LD) representation [35], which has been widely used in realistic systems. In this formalism, the adiabatic basis at time t is used to construct the LD basis for any time τ during the time interval $[t, t + \Delta t]$,

$$|\phi_i^{\text{LD}}(\mathbf{r}, \tau)\rangle = |\phi_i(\mathbf{r}; \mathbf{R}(t))\rangle, \quad (11)$$

and the LD Hamiltonian matrix includes only diagonal elements. The wave function is expressed as

$$|\psi(\mathbf{r}, \tau)\rangle = \sum_i c_i^{\text{LD}}(\tau) |\phi_i^{\text{LD}}(\mathbf{r}, \tau)\rangle. \quad (12)$$

Then the wave function coefficients $\{c_i^{\text{LD}}\}$ can be easily propagated through

$$c_i^{\text{LD}}(t + \Delta t) = c_i^{\text{LD}}(t) \exp \left[\frac{E_i(\mathbf{R}(t))}{i\hbar} \Delta t \right], \quad (13)$$

where $c_i^{\text{LD}}(t) = w_i(t)$. Finally, we convert the coefficients to the adiabatic representation at time $t + \Delta t$ by

$$w_i(t + \Delta t) = \sum_j c_j^{\text{LD}}(t + \Delta t) S_{ji}^*(t, t + \Delta t). \quad (14)$$

Here, $S_{ij}(t, t + \Delta t) = \langle \phi_i(\mathbf{r}; \mathbf{R}(t)) | \phi_j(\mathbf{r}; \mathbf{R}(t + \Delta t)) \rangle$ is the overlap matrix between the adiabatic states at time t and $t + \Delta t$, which can be computed from the transformation matrix $\mathbf{U}$, $S_{ij}(t, t + \Delta t) = \sum_m U_{mi}^*(\mathbf{R}(t)) U_{mj}(\mathbf{R}(t + \Delta t))$. Due to the high efficiency of the algorithm, we adopt the same time-step size for both nuclear dynamics and wave function propagation in the present study.

2.4. FCC-GFSH

FCC-GFSH carries out state tracking for all adiabatic states at each time step and thus gives correct hopping pathways

when multiple trivial crossings are present within a time step [25, 26]. The continuity between adiabatic states at time t and $t + \Delta t$ can be determined by their overlaps, $\{S_{ij}(t, t + \Delta t)\}$ [17]. A typical state tracking can be performed to find state i' at time $t + \Delta t$ which is maximumly overlapped with state i at time t ,

$$i' = \underset{j}{\operatorname{argmax}} [|S_{ij}(t, t + \Delta t)|]. \quad (15)$$

In this study, we aim to find a permutation of all the states at time $t + \Delta t$, $\{f(i)\}$ ($i = 1, 2, \dots, N$), which can maximize the absolute value of trace of the overlap matrix $\mathbf{S}$, $Z_{\max} = \sum_i |S_{i, f(i)}(t, t + \Delta t)|$. Such a permutation can be obtained by the Min-cost algorithm [17] or the Kuhn–Munkres algorithm [36] implemented within SPADE. Note that this approach is much more reliable than the traditional energy ordering for state tracking [1].

Compared with FSSH, GFSH shows better performance in large systems with complex surface crossings because it relies merely on the quantum populations. Following GFSH, we calculate the population changes through [27]

$$\Delta P_i = |w_{f(i)}(t + \Delta t)|^2 - |w_i(t)|^2. \quad (16)$$

We classify all states into a group A and a group B with reduced and increased populations, respectively. The total hopping probability from the active state a to all other adiabatic states during a time step is

$$g_{\text{tot}}(t) = -\frac{\Delta P_a}{|w_a(t)|^2}. \quad (17)$$

If $g_{\text{tot}}(t) > 0$, $a \in \text{A}$ and the hopping probability to state $f(i)$ at time $t + \Delta t$ is defined as

$$g_{a \rightarrow i}(t) = \begin{cases} g_{\text{tot}}(t) \frac{\Delta P_i}{-\sum_{j \in \text{A}} \Delta P_j} & (\text{if } i \in \text{B}) \\ 0 & (\text{if } i \in \text{A and } i \neq a) \end{cases}. \quad (18)$$

Otherwise, if $g_{\text{tot}}(t) \leq 0$, all $g_{a \rightarrow i}(t)$ ($i \neq a$) are set to zero. Then, a uniform random number $\zeta \in [0, 1]$ is generated, and the surface hop to state $f(b)$ should occur if $\sum_{i=1}^{b-1} g_{a \rightarrow i}(t) < \zeta \leq \sum_{i=1}^b g_{a \rightarrow i}(t)$ ($b \neq a$). Instead of adjusting the nuclear velocities after a surface hop [1], a Monte Carlo scheme is utilized for the statistical description of energy conservation [37, 38]. Another uniform random number η is generated. If $\eta < \exp[-(E_{f(b)} - E_a)/k_B T]$, the hop to state $f(b)$ at time $t + \Delta t$ is accepted; Otherwise, the hop is cancelled, and the active state is reassigned as $f(a)$.

2.5. Decoherence correction

A quantum system in contact with a classical bath tends to lose coherence gradually. Due to the classical treatment of nuclei,

however, the decay of coherence is missing in the traditional TSH [1]. As a result, a proper decoherence correction is generally needed [39–54]. In this study, we use a simplified energy-based decoherence time formula to calculate the decoherence time for a general non-active adiabatic state i [20],

$$\tau_i = \frac{\hbar}{|E_a - E_i|}. \quad (19)$$

At each time step, after the electronic propagation, the wave function coefficients of non-active states are rescaled according to

$$w'_{i(\neq a)} = w_i \exp(-\Delta t / \tau_i), \quad (20)$$

and the wave function coefficient of the active state is then renormalized as

$$w'_a = \frac{w_a}{|w_a|} \sqrt{1 - \sum_{j \neq a} |w'_j|^2}. \quad (21)$$

2.6. Calculation of charge mobility and delocalization strength

We focus primarily on the carrier mobility and the inverse participation ratio (IPR). The mean squared displacement (MSD) is calculated by [55]

$$\text{MSD}(t) = \frac{1}{N_{\text{traj}}} \sum_{l=1}^{N_{\text{traj}}} \langle \psi^l(\mathbf{r}, t) | (\hat{\mathbf{r}} - \mathbf{r}_0)^2 | \psi^l(\mathbf{r}, t) \rangle. \quad (22)$$

Here, N_{traj} is the total number of surface hopping trajectories, $\psi^l(\mathbf{r}, t)$ is the time-dependent electronic wave function along the l th trajectory, $\hat{\mathbf{r}}$ is the Cartesian coordinate operator, and $\mathbf{r}_0 = \sum_{l=1}^{N_{\text{traj}}} \langle \psi^l(\mathbf{r}, t) | \hat{\mathbf{r}} | \psi^l(\mathbf{r}, t) \rangle / N_{\text{traj}}$ is the average charge position. When $\text{MSD}(t)$ reaches a linear regime, equilibrium diffusion is achieved. Then, the charge mobility can be computed through the Einstein relation,

$$\mu = \frac{e}{2nk_B T} \lim_{t \rightarrow \infty} \frac{d\text{MSD}(t)}{dt}, \quad (23)$$

where e is the elementary charge. N is the dimension of the system, which is set to be 2 in the present study. The IPR is adopted to measure the spatial size of the electron,

$$\text{IPR}(t) = \frac{1}{N_{\text{traj}}} \sum_{l=1}^{N_{\text{traj}}} \frac{1}{\sum_i |\langle \chi_i | \psi^l(\mathbf{r}, t) \rangle|^4}. \quad (24)$$

It is apparent that the IPR equals to one if the electron is completely localized on a single diabatic state.

3. Results and discussion

3.1. Time-step size and system size convergence

We first investigate hexagonal molecular crystals with $V_a = V_b = V_c = V = (10; 50; 200) \text{ cm}^{-1}$ at 300 K. In figure 2(A), we show the carrier mobilities obtained by different time-step sizes. For each calculation, the total simulation time is 500 fs. Taking the results with $\Delta t = 1$ fs as benchmarks, the maximum error is less than 3% and 17% with $\Delta t = 2$ fs and $\Delta t = 5$ fs, respectively. Considering the large mobility difference for different electronic couplings, we could safely use $\Delta t = 5$ fs to get satisfied description of the charge transport. Thereby, we use this large time-step size in the following studies to achieve a good balance between accuracy and efficiency.

In figure 2(B), we further study the system size dependence of mobility with $\Delta t = 5$ fs. As the number of molecular sites increases, the mobility rises first and then stabilizes to a converged value, indicating that size-independent results have been obtained. Due to the frequent diagonalization of electronic Hamiltonian matrices, minimizing the system size as much as possible is of significance to reduce computational costs. Thereby, in the following studies, $N = 631, 1261$, and 6487 are utilized to study the dynamics for systems with $V = (10; 50; 200) \text{ cm}^{-1}$, respectively.

3.2. Role of electronic coupling signs on charge transport

Based on the translational symmetry of crystals, there are three independent electronic couplings V_a , V_b , and V_c in hexagonal systems and each coupling may take either a positive (+) or a negative (−) value (see figure 1). As illustrations, we take $|V_a| = |V_b| = |V_c| = 50 \text{ cm}^{-1}$ with eight different sign combinations labelled by +++, ---, --+, -+-, +-+, ++-, -+-, and +--+. As shown in figure 3, there is an obvious mobility difference between the systems with $V_a V_b V_c > 0$ (red points) and $V_a V_b V_c < 0$ (green points), indicating that the electronic coupling signs have a strong impact on charge transport. Namely, under the condition of $|V_a| = |V_b| = |V_c| = V$, the charge mobility relies only on the sign of $V_a V_b V_c$. This phenomenon can be explained based on the fact that the electronic Hamiltonian remains identical if any two of V_a , V_b , and V_c are interchanged or the signs of them are changed simultaneously, which was also mentioned by Fratini and co-workers [31]. For simplicity, we focus on two representative sign combinations of V_a , V_b , and V_c (i.e. +++ and ---) below. Note that when V_a , V_b , and V_c take distinct absolute values, the charge mobility is not determined merely by the signs of electronic couplings.

In figures 4(A) and (B), we further investigate the temperature dependence of charge mobility and delocalization strength. In general, when $V > 0$, the charge mobility of systems with weak electronic couplings (e.g. $V = 10$ and 50 cm^{-1}) increases

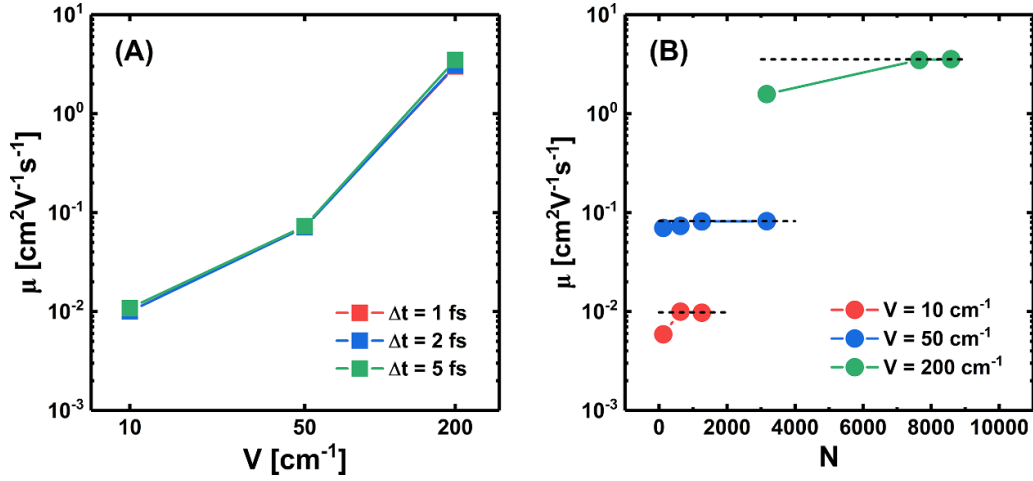


Figure 2. Carrier mobilities of two-dimensional hexagonal molecular systems with (A) $\Delta t = 1, 2$, and 5 fs for different nearest-neighbour electronic couplings and (B) $V = (10; 50; 200)$ cm⁻¹ for different system sizes. To get converged results, the number of trajectories ranges from 1000 to 15 000. The smaller the electronic couplings, the more trajectories are taken. The temperature is set as 300 K and the total evolution time is 500 fs. $\Delta t = 5$ fs gives the highest efficiency while holding a good accuracy. The minimum system sizes for systems with $V = (10; 50; 200)$ cm⁻¹ correspond to $N = 631, 1261$, and 6487, respectively.

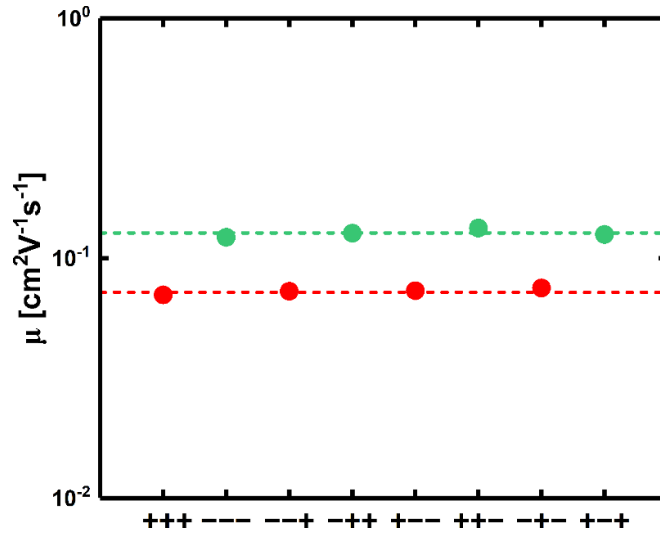


Figure 3. Carrier mobilities for hexagonal molecular systems with different combinations of electronic coupling signs. $|V_a| = |V_b| = |V_c| = 50$ cm⁻¹ is considered. To get converged results, a system with 1261 molecular sites and a time-step size of 5 fs are utilized with 15 000 trajectories. The temperature is set as 300 K and the total evolution time is 500 fs. The red points correspond to the set with $V_a V_b V_c > 0$ and the green points are those with $V_a V_b V_c < 0$. The calculated mobilities have only two values associated with the sign of $V_a V_b V_c$.

first and then decreases as the temperature rises, which meets the traditional hopping mechanism of charge transport. For systems with $V = 200$ cm⁻¹, the temperature dependence of mobility is weaker. It is known that this corresponds to an intermediate mechanism between hopping and band-like transport [26]. In this case, the delocalization strength cannot be well described by the average IPR due to the coexistence of both localized and delocalized states.

Although the transition from hopping to band-like transport has been observed in one- and two-dimensional molecular systems [26], the charge transport in systems with negative electronic couplings has not been systematically studied with surface hopping simulations before. As shown in figure 4(A), the charge mobilities in most systems with negative electronic couplings are larger than those of positive ones with the same absolute values, which is caused by the stronger delocalization

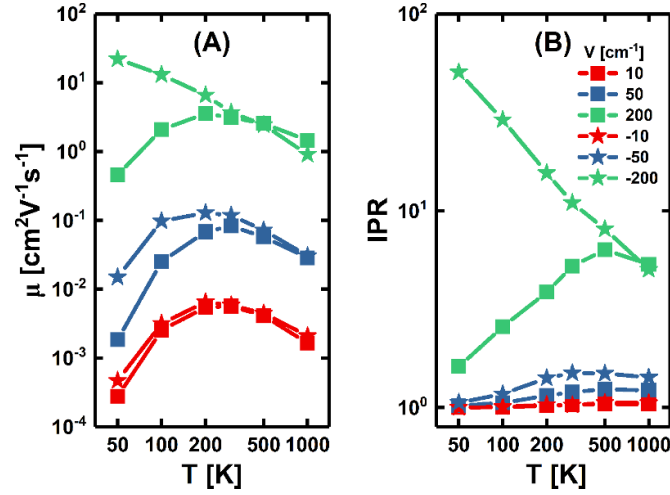


Figure 4. Temperature-dependent (A) charge mobility and (B) average IPR in hexagonal molecular systems with $V_a = V_b = V_c = V = (10; 50; 200; -10; -50; -200) \text{ cm}^{-1}$. To get converged results, the number of trajectories ranges from 1000 to 15 000 according to the magnitudes and signs of the electronic couplings. The smaller the electronic couplings, the more trajectories are carried out. The total simulation time is 500 fs in each calculation.

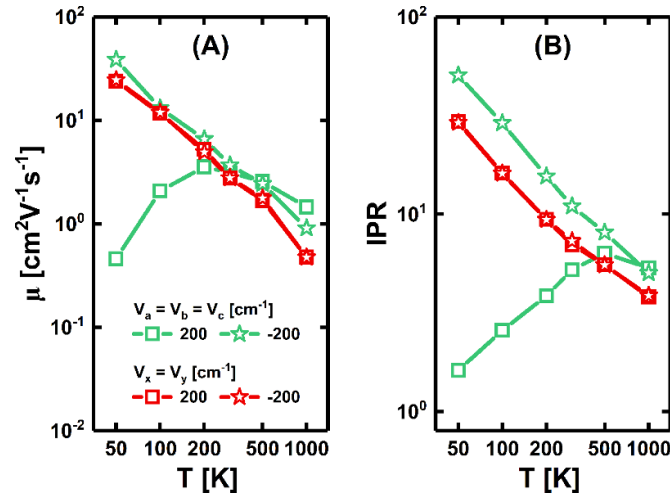


Figure 5. Temperature-dependent (A) charge mobility and (B) average IPR in the hexagonal molecular systems with $V_a = V_b = V_c = \pm 200 \text{ cm}^{-1}$ and square molecular systems with $V_x = V_y = \pm 200 \text{ cm}^{-1}$. To get converged results, the number of trajectories is 2000 and the total simulation time is 500 fs in all calculations. The system size is adjusted to ensure the convergence.

strength of the charge as shown in the temperature-dependent IPR in figure 4(B). As expected, in systems with weak electronic couplings (e.g. 10 and 50 cm^{-1}), the opposite signs of electronic couplings do not significantly affect the charge mobility because only localized diabatic states participate in the hopping dynamics and the charge transfer rate is proportional to the square of electronic coupling [6]. As the electronic couplings change from $V = 200 \text{ cm}^{-1}$ to -200 cm^{-1} , there exist sudden and big changes of the charge mobility. In particular, for $V = -200 \text{ cm}^{-1}$, the charge transport turns to be band-like, where the charge mobility drops continuously as

the temperature rises. The average IPR in figure 4(B) shows a consistent tendency with the charge mobility.

3.3. Comparison of hexagonal and square systems

We also consider square molecular crystals [26], where the sites are arranged in a square lattice and the electronic couplings along the x and y directions are V_x and V_y , respectively. As shown in figures 5(A) and (B), compared with the hexagonal systems studied above, the square systems have significantly different behaviours when changing the signs

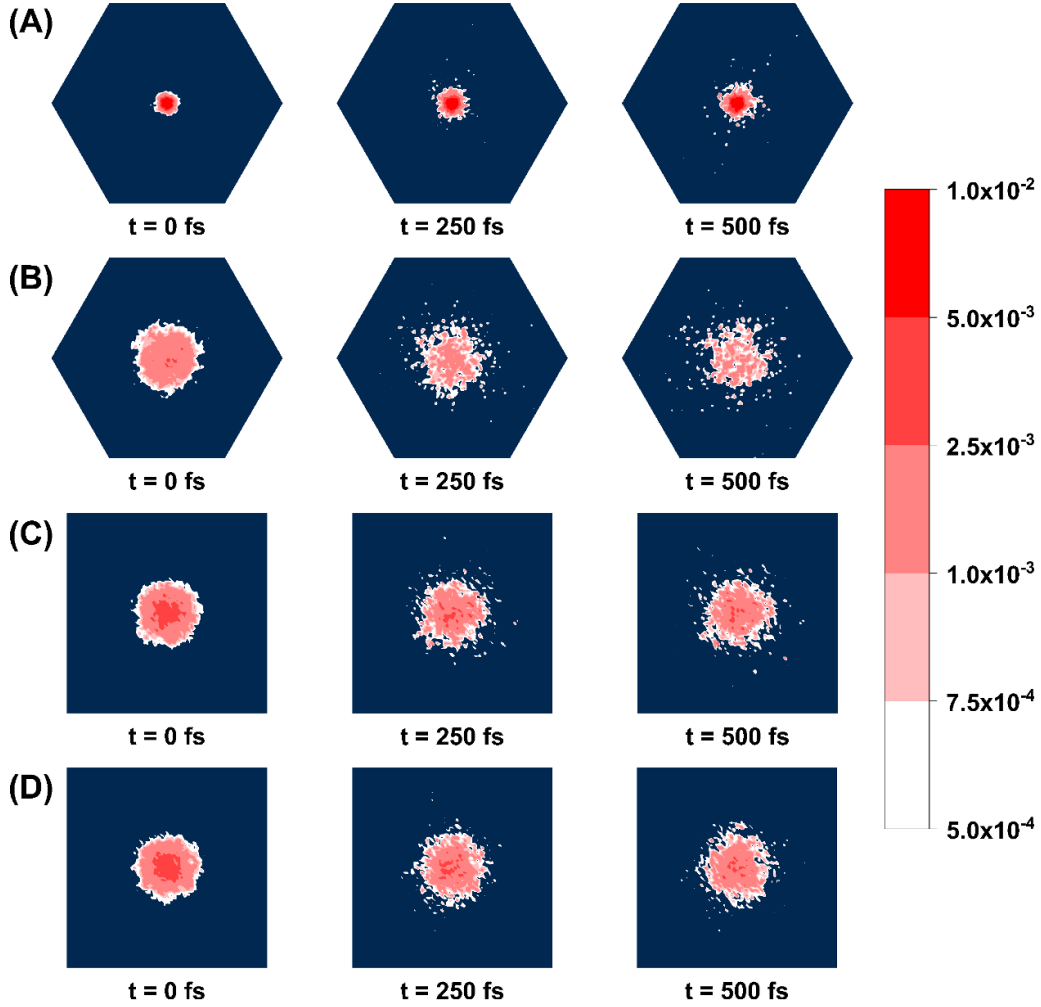


Figure 6. Average spatial distribution of the active states at $t = 0, 250$, and 500 fs in hexagonal molecular systems with (A) $V_a = V_b = V_c = 200 \text{ cm}^{-1}$ and (B) $V_a = V_b = V_c = -200 \text{ cm}^{-1}$ and square molecular systems with (C) $V_x = V_y = 200 \text{ cm}^{-1}$ and (D) $V_x = V_y = -200 \text{ cm}^{-1}$. To get reliable population distributions, the number of trajectories is 2000 in all calculations. The temperature is set as 300 K . The system size is $N = 6487$ in (A), (B) and $N = 6561$ in (C), (D). The charge is initially placed at the centre of the system. The red colours represent the electronic populations with different values. The darker the red colour, the larger the population. Blue represents the population below 0.0005 .

of electronic couplings. Namely, the charge mobilities and delocalization strengths with negative electronic couplings are exactly the same as those with positive electronic couplings in square systems.

In figure 6, we show the average spatial distribution of active states for hexagonal systems with $V_a = V_b = V_c = V = 200$ and -200 cm^{-1} and square systems with $V_x = V_y = 200$ and -200 cm^{-1} in representative snapshots (i.e. $t = 0, 250$, and 500 fs). For hexagonal systems with $V = 200 \text{ cm}^{-1}$, the charge spreads over a small region, while in the systems with negative electronic couplings, the charge is much more extended (see figures 6(A) and (B)). In square systems, the charge distribution is nearly consistent regardless

of the signs of electronic couplings, both with extended charges participating in the charge transport (see figures 6(C) and (D)).

To reveal the underlying mechanism, we investigate the density of states (DOS), energy distribution of active states ($P(E_a)$), and the energy-resolved average IPR. Figure 7 gives the results for hexagonal systems with $V_a = V_b = V_c = V = 200$ and -200 cm^{-1} and square systems with $V_x = V_y = 200$ and -200 cm^{-1} . For hexagonal systems, all the three considered energy distributions differ for $V = 200$ and -200 cm^{-1} , which explains the difference on the charge mobility and the transition from intermediate to band-like charge transport mechanism in the two cases. Namely, the

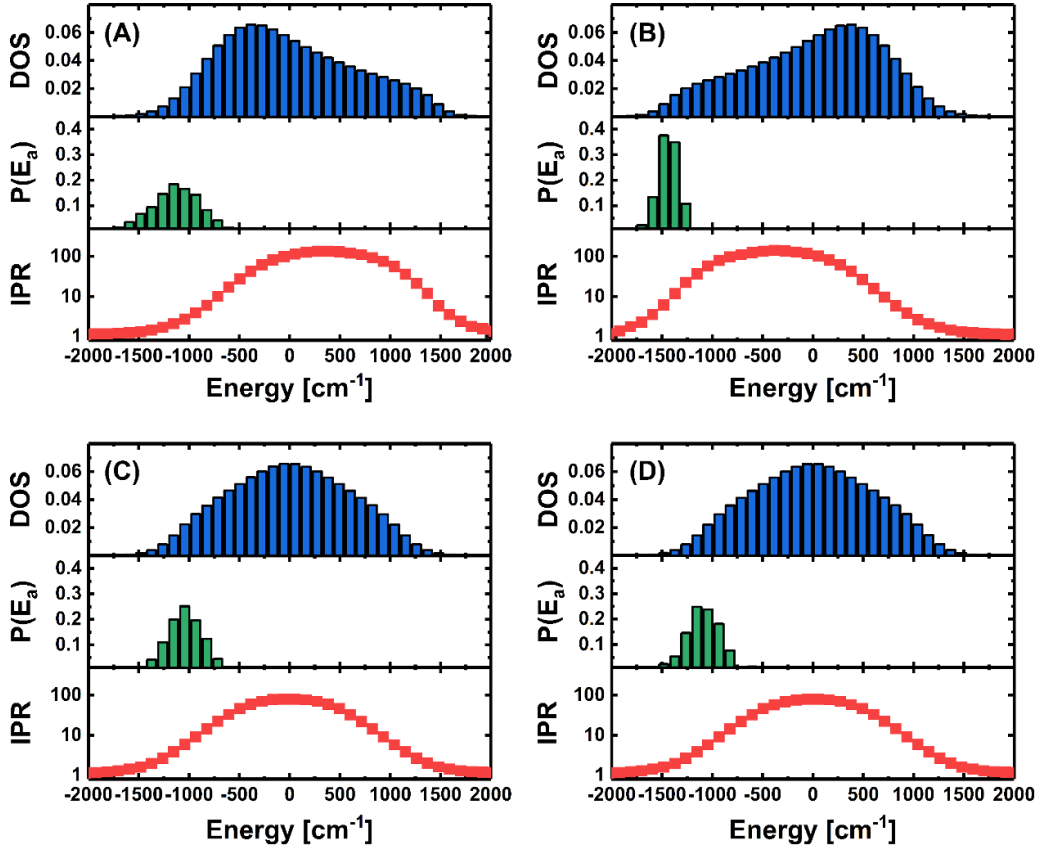


Figure 7. Density of states (blue columns), energy distribution of active states (green columns), and energy-resolved average IPR (red squares) in hexagonal systems with (A) $V_a = V_b = V_c = 200 \text{ cm}^{-1}$ and (B) $V_a = V_b = V_c = -200 \text{ cm}^{-1}$ and square systems with (C) $V_x = V_y = 200 \text{ cm}^{-1}$ and (D) $V_x = V_y = -200 \text{ cm}^{-1}$. To get reliable population distributions, the number of trajectories is 2000 in all calculations. The temperature is set as 300 K and the total evolution time is 500 fs. The system size is $N = 6487$ and 6561 in hexagonal and square systems, respectively. The charge is initially located at the centre of the system.

electron is localized on no more than 10 molecules in systems with $V = 200 \text{ cm}^{-1}$, resulting in localized and delocalized states both involved in the dynamics. For systems with $V = -200 \text{ cm}^{-1}$, the electron is mainly delocalized at larger sizes ranging from 10 to 100 molecular sites. These phenomena are related to the asymmetric DOS and changing electronic coupling signs reverse the DOS completely (see figures 7(A) and (B)). Similar to the population distribution in square systems, the DOS and IPR keep exactly the same with positive and negative electronic couplings (see figures 7(C) and (D)), which agrees with their consistent mechanism of charge transport.

In particular, we find that the energy distribution of active states does not satisfy the Boltzmann distribution in both hexagonal and square systems. In figure 8, we show the spatial distribution of representative states with energies lower than that of the active state. It is apparent that these low-energy states are distributed close to the edge of the systems. In the

nonadiabatic dynamics simulation of charge transport, the carrier is initially placed at the centre of the system and thus cannot reach the edge directly. In fact, during equilibrium diffusion, a large enough system should be utilized to eliminate the edge effect. Therefore, few states in the low-energy region are explicitly involved in our simulations of charge transport dynamics. Note that these edge states can be fully eliminated with periodic boundary conditions and then the energy distribution of active states should follow the Boltzmann distribution.

Finally, we note that the intermolecular electronic coupling is sensitive to the distance and orientation of molecules. Especially, it has been shown that the electronic coupling shows oscillations when shifting one molecule with respect to the other along both transverse and longitudinal axes in a pentacene dimer [10]. Thereby, the signs of electronic couplings can be tuned through molecular design and change the intrinsic charge transport mechanism.

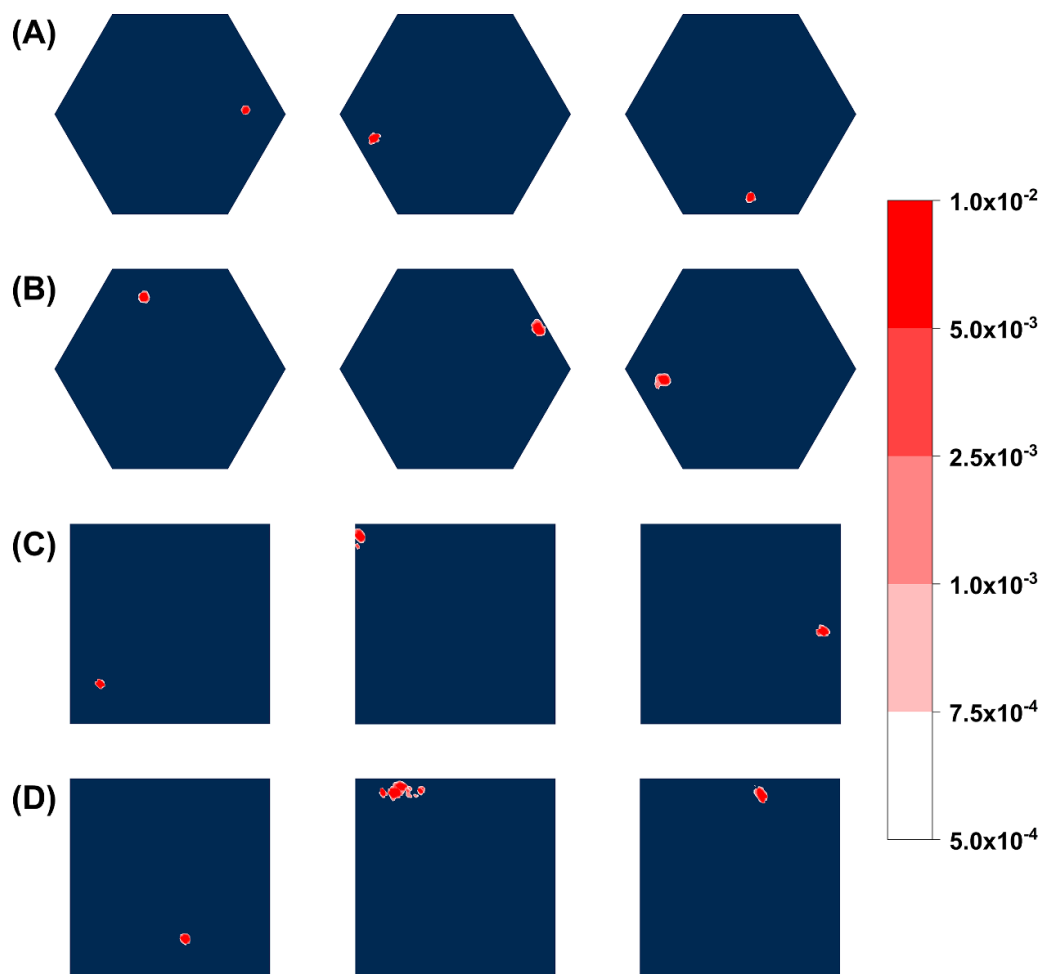


Figure 8. Spatial distribution of the representative states with lower energies than the active state in hexagonal molecular systems with (A) $V_a = V_b = V_c = 200 \text{ cm}^{-1}$ and (B) $V_a = V_b = V_c = -200 \text{ cm}^{-1}$ and square molecular systems with (C) $V_x = V_y = 200 \text{ cm}^{-1}$ and (D) $V_x = V_y = -200 \text{ cm}^{-1}$. For each simulation, the temperature is set as 300 K and the total evolution time is 500 fs. The system size is $N = 6487$ in (A), (B) and $N = 6561$ in (C), (D). The charge is initially located at the centre of the system. The various colours represent the electronic populations with different values. The darker the red colour, the larger the population. Blue represents the population below 0.0005.

4. Conclusions

In summary, we have applied the state-of-the-art surface hopping methods to systematically study charge transport in two-dimensional hexagonal molecular crystals. Fast time-step size and system size convergence have been achieved simultaneously. We have illustrated that the signs of electronic couplings have strong impacts on the charge mobilities in hexagonal molecular crystals. Namely, when all the electronic couplings have the same absolute values, the charge mobilities for systems with negative $V_a V_b V_c$ are larger than those of positive ones. We have shown that adjusting the total sign of $V_a V_b V_c$ destroys the symmetry of the electronic Hamiltonian and changes the energy distribution of active states. Transition from hopping to band-like transport in hexagonal systems has been realized through changing the electronic coupling signs. For comparison, charge transport in square molecular systems does not depend on the signs of electronic couplings. Thereby, we have shown that hexagonal crystals possess an intrinsic contribution from the electronic coupling signs, which could

be used for molecular design of organic semiconductors with required charge transport properties.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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ORCID iD

Linjun Wang  <https://orcid.org/0000-0002-6169-7687>

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